ActPack Gearbox Slip Investigation

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Test Setup & Method

Setup

The ActPack output was coupled through a Futek torque sensor to a fixed plate, so the output was grounded and could not rotate (Fig. 1). With the output grounded, any motor rotation beyond small elastic wind-up means the drivetrain is slipping internally. The motor ran in current control from a Raspberry Pi through the Dephy flexsea API.

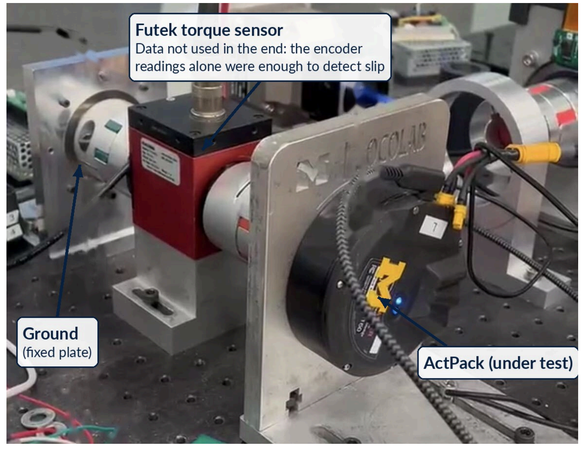


Figure 1. Grounded test setup. Slip was detected from the motor encoder alone; the Futek data were not used in the final analysis.

Ramp test

The commanded output torque ($I_q \times k_t \times N$, with k_t = 0.145 Nm/A and N = 9) was stepped up every 2 s while the motor encoder was logged. The first test used 1 Nm steps up to 15 Nm. The final test used 0.5 Nm steps up to 25 Nm, holding each step above 20 Nm for only 1 s to limit heating at peak current. Slip appears as the motor (input) turning freely while the output stays grounded.

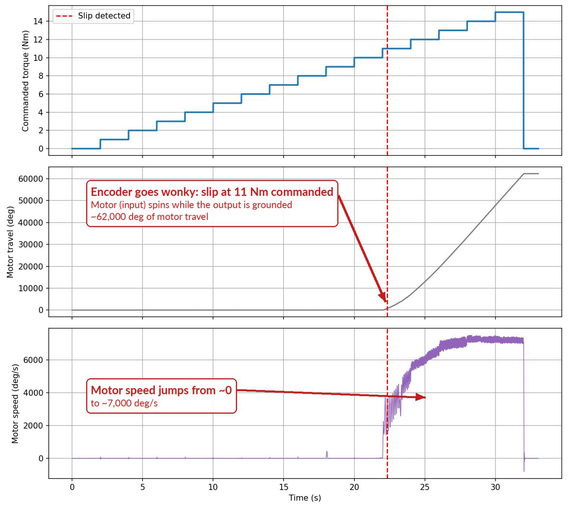


Figure 2. First test (original sun gear). The drivetrain held to 8 Nm, crept at 9 Nm, and slipped fully at 11 Nm commanded, after which the motor turned ~62,000° at up to ~7,000 deg/s.

Disassembly & Diagnosis

The actuator uses a single planetary stage: the motor rotor drives the sun gear, the ring gear is held in the flange, and the planet carrier is the output (Fig. 3). After disassembly, the planets showed no chipped teeth or tool damage, which left two interfaces that could slip under load: the sun gear on its rotor seat and the ring gear in its flange. The ring was the first suspect, because the Gen 2 ring has no anti-rotation key (Gen 1 had one) and relies on its fit alone.

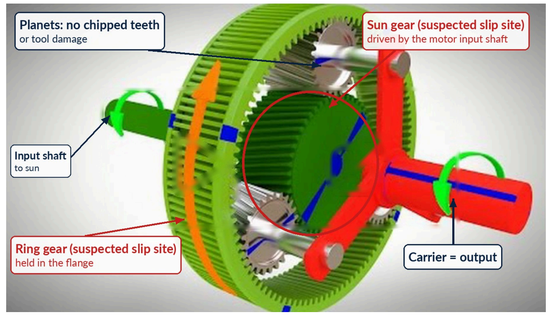


Figure 3. Planetary stage. Red: the two suspected slip sites (sun and ring).

Witness marks

Before reassembly, a Sharpie line was drawn across each suspect joint (Figs. 4 and 5), so that any relative rotation during a slip would show as a broken line.

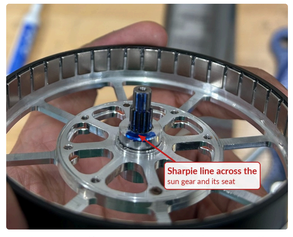


Figure 4. Sun gear marked across its seat.

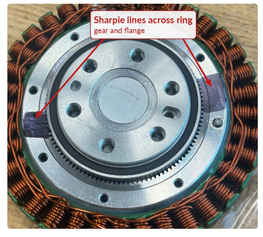


Figure 5. Ring gear marked across the flange.

Re-test

The ramp test was rerun with the marks in place. Slip recurred, and the test was stopped as soon as the slip was heard, so no plot is shown. The motor was then removed again to read the marks.

Second teardown

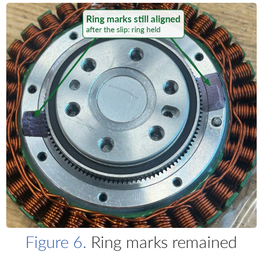


Figure 6. Ring marks remained intact: the ring did not slip.

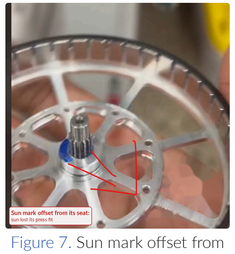


Figure 7. Sun mark offset from its seat.

Root cause: the sun gear lost its press fit on the rotor, so the motor input kept turning while the sun stayed stationary against the grounded output. The ring was not at fault (we first suspected the ring, but the issue was the sun).

Fix, Verification & Safety

Sun gear replaced

With the sun gear replaced, the ramp test was repeated from 0.5 Nm up to the 25 Nm peak (Fig. 8).

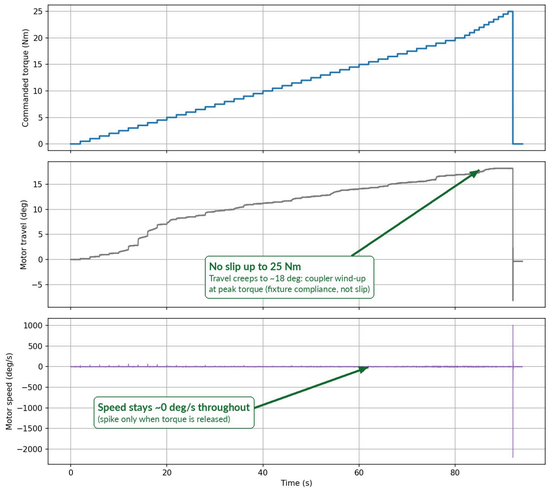


Figure 8. After replacing the sun gear: no slip up to 25 Nm. The ~18° of travel is coupler wind-up at peak torque, as the coupler could not hold the output perfectly rigid.

	Before (Fig. 2)	After (Fig. 8)
Slip	Full slip at 11 Nm	None up to 25 Nm
Motor speed	Up to ~7,000 deg/s	~0 deg/s
Motor travel (max − min)	~62,000°	~26° (−8° to 18°)

Safety protocol

- Check the rotor's permanent magnets for metal shavings before reassembly; the magnets hold ferrous debris, which can rub the windings or enter the gears.
- If a screw is seized in the housing, do not try to remove it: level it flush with the housing and keep all debris out of the motor.
- The Gen 2 ring has no key (Gen 1 had one), so avoid prying or impact loads on the ring, and re-mark the ring and sun after every rebuild.
- Label the three motor phase leads before desoldering them (see Appendix).

Appendix: drive fault (status_ex = 8)

After the phase leads were resoldered in an unknown order, the motor held stiff (~1.15 A flowing) instead of rotating, and running Find Poles then stored a faulty calibration that set status_ex = 8 and blocked all motor current. Lead order matters: only 3 of the 6 orders work (cyclic rotations, e.g., A-B-C, B-C-A, C-A-B), and any single swap reverses the phase sequence. A phase lead touching a battery-voltage component also kept Find Poles from completing. Fix: clear the contact, swap two leads back, run Find Poles from the Dephy GUI (the motor ticks for ~30 s and the output turns ~10° or more), then power cycle.

